Token-decoder gap-filler (io4_tokdec; Job 3 io4_tokdec_blend adds the dashed skip)

chl_cnn/tokdec.py TokDecNet + chl_cnn/models.py DilatedCNN. train_pinn.py --model temporal3d --t3d-stem tokdec --ptok-patch 4 --tokdec-dim 128 --tokdec-ch 32 --tokdec-heads 4 --tokdec-layers 2 [--tokdec-blend] --context 8 --batch 8

B = batch, T = 17 days (t−8 … t+8), H × W = 149 × 181 (0.25° grid), patch grid 38 × 46 (P = 4 px), D = 128, C = 32. io4_tokdec parameters 1,418,787: stem 796,704, backbone 622,083.

patch grid 38 × 46

full resolution 149 × 181

Arrows

- full-resolution pixels
- tokens / patch values
- other tensors
- Job 3 only (blend skip)
- day t marker

1 Inputs per day

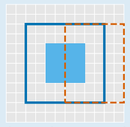
d = t−8 … t+8, 12 channels:
CHL z-anomaly vs DOY
clim. × visible mask
visible mask
9 physics fields (z)
× ocean mask
ocean mask
→ (B, 12, 17, H, W)

context days carry
the donor's clouds

t−8 … t … t+8

all 12 channels

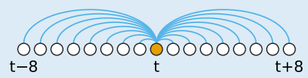
2 ViT patch embedding



4 × 4 patch (filled)
8 × 8 window (blue)
neighbour window,
4-px stride (dashed)

shared across days; pad to 152 × 184
Conv2d(12, 128, kernel 8,
stride 4, padding 2): 8 × 8 windows
on a 4-px stride → 38 × 46
LayerNorm + day-offset emb.
+ patch-position emb.
only ocean patches kept:
→ (B, Np, 17, 128)

3 Temporal self-attention



within each ocean patch,
over its 17 day tokens;
no attention across patches
2 pre-norm blocks, 4 heads,
fused SDPA, batched over
patches: (B · Np, 17, 128)
days missing from the store
masked out of the keys
→ LayerNorm

4 Attention pooling

query: day t's output token
keys / values: all 17 output
tokens, day t included
4 heads → pooled (B, Np, 128)

pooling weights recorded
(by day offset, map
autocorrelation)

concat [pooled, day-t token]
→ (B, Np, 256)

Context days (d ≠ t) reach the backbone only through the tokens:
8 × 8 px windows on a 4-px stride, decoded to 32 channels per 4 × 4 patch.

Day t also enters at full resolution through its own channels (6).
No per-day blend, no FiLM.

Job 3 adds each context day's own visible pixel values
through the blend (dashed), as in io3_ptok4.

Job 3 only (io4_tokdec_blend): io3 arm-A blend from the same output tokens

patch score s = W_q tok(t) · W_k tok(d) / √128, bilinear between ocean-patch centres → (B, 17, H, W)
+ per-pixel term (1×1-conv MLP on visible$_d$, |d−t|/8, (d−t)/8, |SST$_d$ − SST$_t$|)
softmax over context days d ≠ t visible at p + learned "no day" logit
blend = Σ$_d$ w$_d$(p) a$_d$(p) with a$_d$ = own visible CHL anomaly; mass = 1 − w$_{none}$; entropy → 3 extra channels

5 Decoder

Linear 256 → 512 per patch
= 4 × 4 px × 32 ch
pixel shuffle → (B, 32,
152, 184), land patches 0
crop to 149 × 181
Conv 3×3 32 → 32 + GELU
(smooths patch seams)
→ (B, 32, H, W)

6 Backbone input

concat at full res:

decoded
context (32)

day t's own
channels (11)

static (7)

→ (B, 50, H, W)
Job 3: + 3 → 53

7 Dilated backbone

Conv 50 → 64, 3×3 (no FiLM)
8 residual blocks, width 64
x + [BN, GELU, Conv 3×3, dilation d] × 2
d = 1,2,4,8,16,1,2,4
BN, GELU, Conv 64 → 3, 1×1

output (B, 3, H, W): z of t−1, t, t+1
+ DOY clim z (residual) × sd + mean → ln CHL

loss: L1 day-t hidden px + 0.1 × visible px
+ 100 × gradient term + 1 × L1 t±1 donor-hidden

day t's own channels (B, 11, H, W): physics, visible CHL z, visible mask

static maps, day t (B, 7, H, W):
clim z, lat, lon, log10 depth,
DOY sin/cos, year

static maps (B, 7, H, W)

Cost (median s / epoch over the run's epochs; peak GPU memory allocated)

	io4_tokdec	io4_tokdec_blend	io3_ptok4
stem parameters	796,704	829,826	201,762
backbone parameters	622,083	623,811	605,379
total parameters	1,418,787	1,453,637	807,141
batch	8	8	8
s / epoch	97	108	80
peak GPU memory	11.0 GB	11.0 GB	9.7 GB